iCook: a Smart Tangible Cooking Interface Integrating TUIO Markers, Hand Gestures, Object Detection, Emotion Analysis, and Eye Gaze Tracking

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***Abstract*—User engagement in smart cooking classrooms is a crucial factor in enhancing the user experience and simplifying the cooking process. This study compares Tangible User Interfaces (TUI) and gesture recognition combined with object detection to assess their interactivity and usability in future cooking environments. Eleven participants (aged 18 to 22) prepared selected meal recipes using either TUI or gestures with object-based interactions. Results showed that TUI significantly reduced task completion time (t = -3.2185, p = 0.0053) compared to gesture However, both methods contributed to increased engagement and enjoyment of the cooking experience. Facial expression analysis enabled the evaluation of users' emotional states during cooking. The findings reveal that TUI primarily elicited positive emotions, such as happiness and surprise, whereas gesture-based interaction was more frequently associated with errors and moments of surprise. These results highlight the potential of TUI in optimizing user experience while also demonstrating the unique advantages of both interaction modalities in smart cooking environments.**

***Index Terms*—Interaction design, Tangible User Interface, TUIO, gesture recognition, object detection, eye gaze, facial expression, facial recognition, smart cooking, DollarPy, mediapipe, deepface, YOLO, Dlib.**

I. Introduction

The integration of advanced interactive technologies into daily practices has transformed Human-Computer Interaction (HCI), reshaping how users engage with systems across various domains. This shift has sparked growing interest in enhancing culinary experiences through innovative interfaces. The culinary domain, in particular, presents a unique challenge: integrating physical interactions with intelligent digital systems to support both the learning process and the execution of cooking-related tasks.

This paper presents an HCI system for interactive cooking classes that incorporated Tangible User Interfaces (TUIO), hand gesture recognition, object detection, emotion analysis, gaze tracking, and dynamic contextual content. The primary goal of this system was to create an immersive, adaptable, and user-centered cooking environment that dynamically responds to users' actions and preferences.

Several challenges exist in smart cooking environments, including facilitating multitasking and improving user experience through intuitive interfaces. First, TUIO markers can serve as physical representations of virtual recipe elements, ingredients, and tools. Research on HCI in culinary settings has highlighted how tangible interfaces enhance engagement and effectiveness [1], [2]. However, the integration of emotion analysis and eye-gaze tracking with TUIO is still an emerging area.

Another interaction methodology is the use of hand gestures in smart environments, which offers a more intuitive mode of interaction [3]. Hand gestures provide natural responsiveness, serving as an intuitive alternative to TUIO. Users can select ingredients, progress through steps, and interact with tools through simple gestures, eliminating the need for complex tools or menus. This enhanced accessibility and usability without interrupting the cooking process, as demonstrated in studies on gesture-enhanced cooking instructions [4].

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Furthermore, advanced object detection models such as YOLOv11 enable real-time recognition of ingredients, utensils, and other kitchen elements. However, the combination of object detection with hand gestures and multimodal interaction is still under development to create systems that react dynamically and simulate real-world kitchen scenarios [5]. With this in mind, the integration of object detection with hand gestures can enhance overall control, flexibility, and error reduction in applications such as selection and manipulation [6].

Personalization in such systems must be enhanced to maintain user engagement. For instance, an intelligent recipe-picking function that adjusts meal options based on the time of day can ease users' decision-making [7]. Additionally, incorporating eye-gaze tracking to analyze user behaviors and preferences can generate heat maps, providing insights into improving design and functionality. These heat maps help address usability challenges and optimize user experience [5]. As human-artificial intelligence collaboration advances, future systems must provide even more personalized experiences [8]

Another challenge is the lack of large-scale datasets for human gesture recognition, kitchen ingredients, and tools, which are crucial for improving the accuracy of gesture and object recognition systems [9]. In addition, the integration of emotion evaluation, eye-gaze tracking, and context-aware content adaptation can make modern platforms essential for HCI and affective computing [5]. Researchers and developers are striving to create culinary systems that not only facilitate cooking but also transform it into an engaging and interactive experience [10].

This research contributes in emerging technologies which can be seamlessly integrated into traditional practices like cooking, yielding innovative solutions such as emotion analysis and gaze tracking that enhance both functionality and user experience [11]. The tools employed to assess cooking quality underscore the critical role of accurate feedback in culinary contexts [12] and emphasize the necessity of optimizing home cooking practices through smart cooking systems [13]. Ultimately, these technologies engage users in culinary activities while ensuring that learning outcomes are effectively achieved in an enjoyable and interactive manner.

II. Related Work

The field of smart culinary practices has seen significant growth in recent years. Extensive research has been conducted on various aspects of these technologies, yielding valuable insights into their potential to enhance the cooking experience. These findings provide a foundation for the proposed interactive smart cooking system.

Jelodar et al. [14] conducted a notable study on identifying the states of objects in cooking images. They proposed a deep learning model based on the ResNet architecture to classify cooking items according to their states—such as sliced, diced, or whole. This research is particularly relevant to the proposed framework, as it not only addresses the challenge of identifying culinary objects but also recognizes the transformations they undergo during cooking. Equally important, YOLO—particularly version 11—has demonstrated the ability to operate effectively in real-time, even in complex environments, making it highly suitable for critical applications requiring both high precision and speed [15]. Similarly, Lilja and Piirainen-Marsh [4] explored the role of gestures in multimodal cooking instructions, with a specific focus on second-language cooking classes. Their findings indicate that gestures significantly enhance cooking instructions by making them more interactive, thereby improving comprehension and facilitating the execution of complex tasks [4]. While existing research highlights the benefits of gestures in increasing engagement and understanding, much of it is centered on classroom environments. This focus underscores a gap in the integration of gesture-based systems into digital cooking environments, particularly in conjunction with object detection, to enable more spontaneous and intuitive interactions.

Recent advancements in gesture recognition, particularly those utilizing photoplethysmography-based methods, have demonstrated the feasibility of fine-grained activity recognition in smart home environments [16]. In kitchen settings, such systems can detect specific hand gestures to streamline cooking processes, thereby enhancing both control and monitoring capabilities [16]. Multimodal interaction systems that integrate gesture recognition with object identification and verbal communication have further improved user interaction with intelligent cooking systems [5]. Similarly, technologies such as the Z-Ring, which employs bio-impedance sensing, have enhanced gesture recognition and object detection, enabling highly precise control and interaction in cooking interfaces [17]. Another promising approach involves human-robot collaboration frameworks, which facilitate more spontaneous and seamless interaction experiences [5]. Additionally, serverless gesture recognition systems have gained prominence, offering low-cost, privacy-preserving alternatives for smart cooking interfaces. These systems enable real-time gesture recognition while minimizing reliance on cloud services by leveraging on-device processing [18]. Wearable haptic interfaces, such as smart gloves, have further expanded the potential of gesture recognition by providing evaluative feedback on object manipulation in culinary contexts [19]. These devices replicate tactile sensations, thereby enhancing user engagement with cooking tasks. Furthermore, the development of large-scale datasets for human gesture recognition, such as GestureSet, has been instrumental in improving the robustness and adaptability of gesture recognition systems in culinary applications [9].

Krestanova et al. [11] explored the impact of TUI systems in education, gaming, and cooking, emphasizing their ability to enhance engagement through tangible markers and interactive surfaces. Their study demonstrated how TUI facilitates efficient interaction with cooking instructions and ingredient selection [11]. Additionally, they examined a TUIO-based educational system that improved mathematical learning in preschoolers. While their research confirmed the benefits of TUI in enhancing user engagement, it did not address the integration of advanced features like emotion analysis and eye-gaze tracking, which could further enhance adaptability

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in interactive cooking systems.

Eye-gaze tracking and emotion analysis have become essential components in intelligent HCI systems, enabling more adaptive and context-aware user experiences. Dlib, an open-source machine learning library, has been widely used for real-time eye-gaze tracking through its facial landmark detection capabilities [20]. Dlib can estimate gaze direction by identifying key facial points, allowing systems to analyze user focus and engagement levels [21]. This capability is particularly beneficial in interactive cooking environments, where eye-tracking can be used to assess attention to recipe instructions, ingredient selection, and overall user experience. Complementary to gaze tracking, emotion recognition plays a crucial role in understanding user behavior. DeepFace, a deep learning-based facial recognition framework, has demonstrated high accuracy in emotion classification by leveraging convolutional neural networks (CNNs) [21]. DeepFace can provide insights into user satisfaction, frustration, or confusion during cooking tasks by analyzing facial expressions, allowing the system to dynamically adapt instructional content to improve engagement [21]. The integration of eye-gaze tracking with emotion analysis offers a multimodal approach to enhancing smart cooking experiences, paving the way for more personalized and intuitive interfaces in future HCI applications.

These studies emphasize the importance of integrating multi-modal interactions. While previous research has made significant strides in these areas, the proposed system addresses existing limitations by comparing TUIO markers, YOLOv11-based object detection, and real-time hand gesture recognition in relation to emotion analysis and eye-gaze tracking. This integration facilitates the creation of a smart cooking interface that is not only interactive and engaging but also adaptable to the dynamic nature of cooking, offering users a more intuitive and efficient approach to following recipes and participating in culinary activities.

III. System Architecture and Setup

The experimental setup utilized a laptop as the main display for the graphical user interface (GUI). This setup included a laptop camera and an external webcam. However, environmental factors such as lighting conditions and camera angles significantly impacted the system's efficiency. To mitigate these challenges, flexible positioning camera mounts were used to ensure optimal angles, enhancing the system's ability to capture clear and accurate visual data. Additionally, the use of RGB cameras further improved performance by capturing high-resolution color data, allowing the system to adapt to varying lighting conditions and angles for more reliable detection [22].

To enhance both security and personalization, biometric authentication was implemented. This process involved recording the user's facial features and comparing them with a set of pre-registered profiles to grant access to the class. Once authenticated, the user's culinary experience began with selecting a recipe from a time-sensitive menu. Before 12 PM, the menu offered breakfast, with "eggs" as the default option. After 12 PM, the menu changed to lunch, with "pasta and lemonade" as the default selection. This dynamic adjustment ensured that the system recommended meals based on the prevailing time of day. Each recipe was associated with two interaction methodologies that divided the cooking process into manageable tasks.

After selecting a recipe, users were able to choose between the two interaction methods. First, TUIO assigned unique IDs to tools and ingredients, which were tracked using ReactVision Server, allowing users to interact seamlessly with their kitchen environment. They could select ingredients, adjust quantities in real-time, and control virtual kitchen elements such as stove flames. Additionally, the system included intuitive controllers to assist users throughout the cooking process. On the other hand, the hand gesture recognition and object detection approach utilized YOLOv11 for object detection and employed MediaPipe and DollarPy to track hand gestures. This method enabled real-time recognition of selected objects and gestures, allowing users to interact with the system by performing gestures to select ingredients or manipulate kitchen tools. Fig.1 illustrates the system overview.

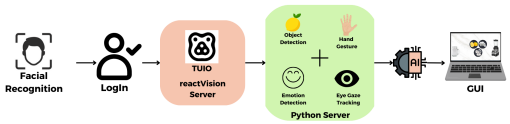

Fig. 1: System Overview

To further enhance user interaction and compare the two methodologies, the system incorporated gaze tracking and emotion recognition technologies. Emotion recognition algorithms analyzed facial expressions to assess user satisfaction, while eye gaze tracking techniques determined users' focus during tasks. The collected data provided valuable insights into the most preferred ingredients, enabling further system personalization. Additionally, this data was utilized to optimize interface placements for future advertising. Therefore, the system improved user personalization, simplified the cooking process, and enhanced overall user satisfaction by leveraging these detailed insights.

IV. Experimental Study

The goal of this study was to evaluate and compare different interactive methodologies within the iCook system, focusing on increasing both usability and user engagement. The study aimed to enhance the interactive learning environment by identifying the most effective and engaging approaches for learners.

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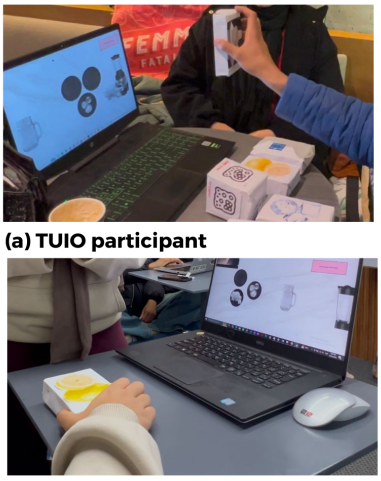

(a) TUIO participant

(b) Object detection and hand gesture participant

Fig. 2: Real-time Experiment Samples

A. Gestures Recognition and Object Detection

Integrating hand gesture recognition and object detection algorithms such as MediaPipe, DollarPy, and YOLOv11; increased user interactivity with smart cooking applications in kitchens. It enabled the camera to detect objects selected by users on crafted cubes labeled with different cooking ingredients and tools used in cooking, such as:

1) **Ingredients:** lemon, ice, water, sugar, egg, oil, salt, pepper, milk, pasta, cheese, butter and chicken.
2) **Tools:** knife, mixer, pan, pot and flame.

Using the YOLOv11 model that was fine-tuned on these crafted cubes, achieving mAP of 99.5 percent and a precision of 96 percent, thereby guaranteeing reliable identification of ingredients and tools. In addition, this approach utilized three dynamic hand gestures:

1) **Selection Gesture:** The user selected a cube to identify the ingredient or tool of interest, while the system recognized both gesture and object.
2) **Move Gesture:** Mapping hand movement in real-time that made users intuitively move the ingredient to any specific place after selecting it.
3) **Back Gesture:** Right to left arrow gesture allowed the user to return to the previous step if something went wrong.

B. Tangible User Interface (TUI)

TUIO was integrated to enable the capabilities of the Tangible User Interface (TUI) and marker provisioning. These markers acted as physical controllers, allowing users to perform important operations on the user interface using physical manipulation. Each ingredient in the cooking class was represented by its own TUIO marker, alongside controllers for actions such as rotate, move and select, similar to those in the gesture recognition section. In Fig.3, some of the crafted cubes and its TUIO ID are shown.

Fig. 3: Objects and its TUIO IDs

C. Face Identification and Expressions

The system incorporated facial recognition and emotion analysis during the login and interaction processes using the DeepFace library. During log-in, the system captured an image from the camera and verified it against stored profiles. If a match was found, the user was granted access; otherwise, the system registered the individual and saved the image with a unique username. After log-in, frames captured from the live camera were continuously fed into the system, which detected facial expressions and stored the results in a CSV file. These expressions were later analyzed to enhance the user experience.

Integrating emotion analysis with TUIOs and object detection driven by hand gestures enabled comparative research, providing insights into user satisfaction and engagement across different interaction methods. The correlation between recognized emotions and specific interactions, such as object selection, hand gesture commands, or TUIO manipulation, helped to determine user preferences. For example, comparing users' emotional responses during TUIO selection versus gesture-based object selection revealed which mode elicited more positive emotions, indicating a more intuitive and enjoyable experience. Additionally, combining emotional data with evaluation metrics such as task completion time and error rates provided a deeper understanding of how different interaction modalities influenced user performance and emotional well-being. Consequently, emotion analysis plays a crucial role in designing user-friendly systems that balance usability and user satisfaction.

D. Real-Time Eye-Gaze Tracking

The eye-gaze tracking system utilized Dlib models for face detection and shape prediction to determine the precise location of facial landmarks and eye positions in real time. The system detected faces using Dlib frontal face detector, then extracted only the landmarks around the eye regions (indices 36 to 47) by systematically recording and saving their X and Y coordinates to a CSV file. The collected gaze data were used

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to generate heatmaps, highlighting areas where users focused their attention on the GUI.

The process involved preprocessing the gaze coordinates to align with the image dimensions, applying Kernel Density Estimation (KDE) to create a smooth distribution of gaze points, and visualizing the data through color heatmaps. These heatmaps facilitated the analysis of user interest by identifying focal areas and suggesting improvements in applications such as human-computer interaction, user behavior analysis, and gaze-based interaction systems. In addition, they were used to evaluate the system interface, determining whether the GUI was cluttered or visually appealing.

Fig.4 presents a heatmap illustrating areas of high user activity for one of the recipes in the iCook system. The intensity of the color ranges from blue, representing low interaction density, to red, indicating high activity. According to the heatmap, users focused most of their attention on the mixer, suggesting frequent use or hand presence in that area. Moreover, high-intensity regions near lemon, ice, and sugar indicate particular user interest in these ingredients. This analysis highlights key activity areas, providing valuable insights into system efficiency and ease of use.

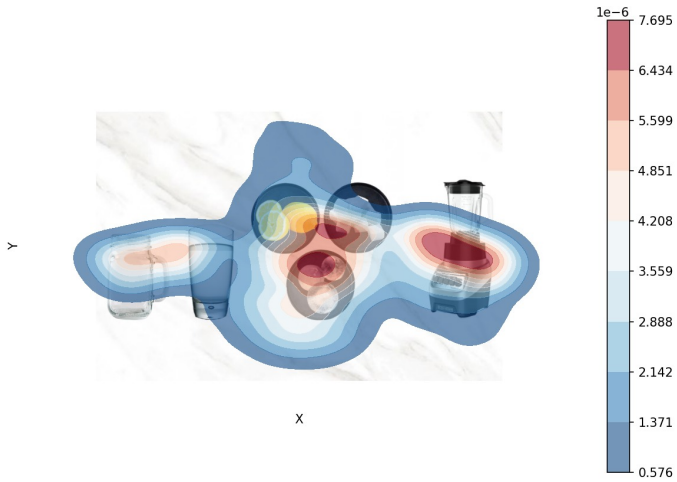

Fig. 4: Heat-map Visualization for Eye Gaze Tracking

E. Participants

The experiments were conducted on a group that consisted of 11 undergraduates between 18 and 22 years, aiming to evaluate the effectiveness of TUIO, Hand Gesture Recognition and Object Detection methodologies. The goal was to determine which methodology resulted in faster learning and usage by the participants.

F. Procedure

The purpose of this experiment was explained to each participant, and they were instructed on how to use the system, including signing up via face detection and rotating markers to select a recipe and one of the two interaction methodologies: TUIO or Hand Gesture Recognition combined with Object Detection. Once the setup was complete, participants began cooking the chosen recipe using their selected methodology, following on-screen notifications until the recipe was finished. After completing the first experiment, participants were required to repeat the process, starting from log-in and proceeding through the recipe without assistance to evaluate usability and learnability. Experimenters were present throughout the process to assist participants when necessary, collect feedback, and record the time taken by each participant to complete the experiment.

G. Results

The statistical analysis revealed a significant difference in task completion times between the TUI and Hand Gestures with Object Detection methodologies. Outliers were identified and removed using the interquartile range (IQR) method, where the first (Q1) and third quartiles (Q3) were calculated, and any data points falling outside the range $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ were excluded, ensuring a more robust statistical analysis. After removing outliers, the 95% confidence interval (CI) for TUI ranged from 46.39 to 87.80 seconds, while the 95% CI for Hand Gestures with Object Detection was 90.76 to 160.15 seconds. The lack of overlap between these intervals strongly suggests that TUI enables significantly faster task completion than gesture-object method. Furthermore, a t-test comparing the two methodologies yielded a t-statistic of -3.2185 and a p-value of 0.0053, indicating that the difference in performance is statistically significant ($p < 0.05$) and unlikely to be due to random variation. The negative t-value confirms that TUI had a lower average completion time than gesture-object. These findings highlight the efficiency of TUI in interactive cooking environments, demonstrating its potential for enhancing usability and reducing task execution time compared to hand gesture-based interactions. In Fig.5, the bar chart clearly illustrated that using TUI approach was faster than using the gestures-object detection approach.

HCI.png

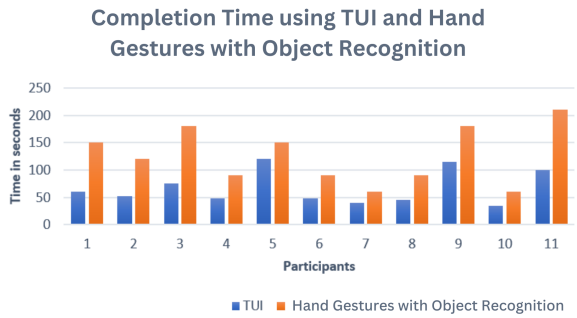

Fig. 5: Completion time for each participant using TUI and Hand Gestures with Object Detection

Emotion analysis revealed that TUI predominantly elicited positive emotions such as happiness and surprise, whereas

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gesture-object tracking often resulted in surprise and occasional frustration due to errors. Furthermore, eye-gaze tracking indicated that most users concentrated their focus on the center of the screen, suggesting that key interface elements were effectively positioned to capture attention. At the end of the experiment, participants provided feedback confirming that the TUI methodology was faster and easier to use than gesture-object approach. However, they also acknowledged that the gesture-object interaction provided an intuitive and engaging experience.

H. Limitations

One of the major limitations of this study was the small sample size of 11 participants in a narrow age range of 18 to 22 years. The system was designed for a wider age group of 10 to 35 years, making it necessary to expand the participant pool to assess its effectiveness across different age ranges. Additionally, participants tried only one recipe, limiting the evaluation of their competence in using the system for various cooking tasks. The training data for the hand gesture recognition system were also small, which may affect the system's ability to generalize across various subjects and cooking scenarios.

V. Conclusion and Future Work

This study compared Tangible User Interface (TUI) and Hand Gestures to Object Detection in the iCook system. Based on eye-gaze tracking and emotion analysis to enhance personalization. Participants selected recipes and performed cooking actions using marker-based interactions (TUI) or hand gestures, providing usability and engagement insights. Results showed that TUI significantly reduced task completion time ($t = -3.2185$, $p = 0.0053$) compared to gestures, with 95% confidence interval confirming its efficiency. While gestures caused some frustration because of recognition errors, emotion analysis showed that TUI had positive emotions like happiness. Eye-gaze data indicated that key interface elements successfully captured user attention. Future work should include a more diverse participant group and multiple recipes to better assess adaptability. To improve recognition accuracy, the hand gesture dataset must be expanded. This study provides a foundation for developing more interactive and personalized cooking systems, increasing user engagement and usability.

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